

CAPSTONE IN COMPUTER SCIENCE I

Shadow

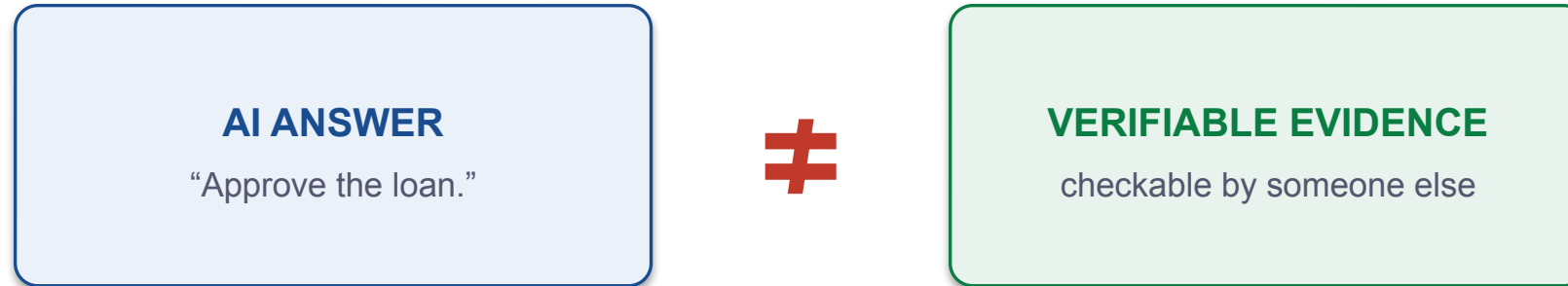
A Cryptographically Verifiable Evidence and Spatial
Audit System for AI-Assisted Decisions

From Multi-Agent Answers to Independently Verifiable AI Decision Evidence

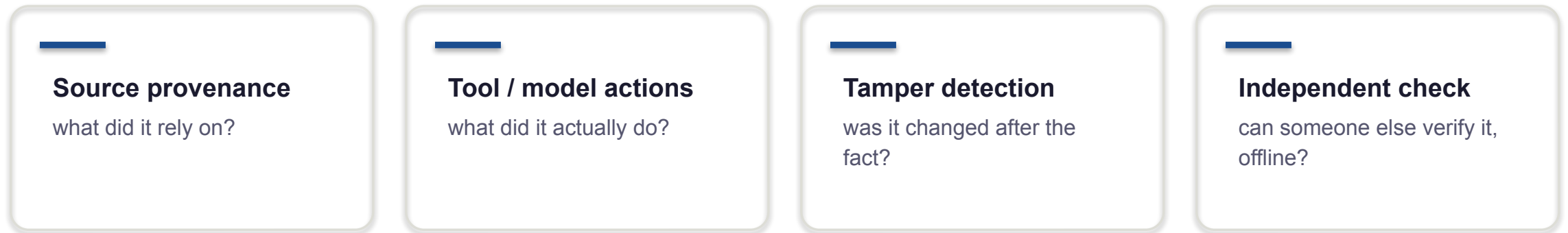
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Practice presentation · 2026-07-21

An AI answer is not verifiable evidence



Ordinary logs are mutable and platform-specific. A model’s explanation is generated prose, not a record.



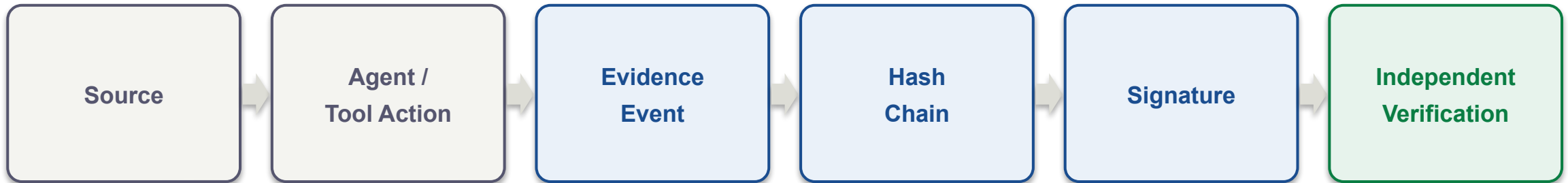
From more opinions to verifiable evidence



Trust moved **down a layer** — to the evidence underneath, not the voices on top.

A refinement of the original research question, not a project that failed.

The evidence lifecycle



- Each event is structured and canonically serialized.
- Each event commits to the previous one — change an early event, everything after it **breaks**.
- The sealed batch root is signed (Ed25519). Anyone can verify the record — **including offline**.

One core, three profiles, many surfaces

CORE

canonical evidence schema · signed hash-chain record · source maps
verifier · claim–evidence graph · ingested-output audit (structural)
host-tested

PROFILES

banking-v1 · data-science-v1 · coding-agent-v1
one verification grammar across domains

INTERFACES

CLI · MCP (11 tools) · HTTP · browser verify.html | Unity Shadow Lens ·
Three.js replay · Android mock
verifier: browser-rendered · Unity/3D: prototypes, device-validation pending

Interfaces are **not** equally mature — the deck labels which is which.

Same machinery across domains

Banking

document → risk claim → source

Data Science

dataset → model → metric

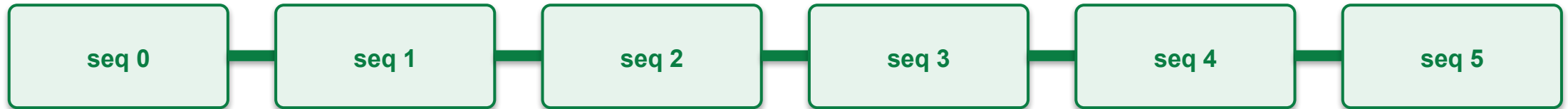
Coding Agent

issue → diff → tests → commit

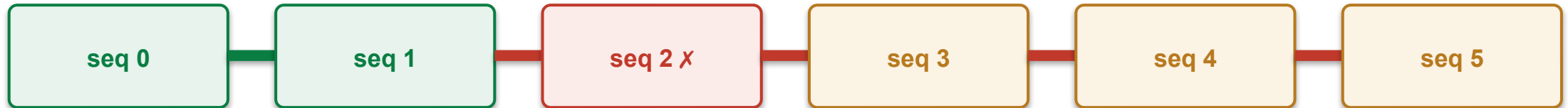
Underneath, the same verification grammar: **sequence** → **hash chain** → **signature** → **source resolution**.

Tamper demonstration

PRISTINE



TAMPERED — one field changed at seq 2 ↑ first failed sequence



The verifier names the **exact first failed sequence** — not “something is wrong.”

Everything after it is **invalidated** — the signature covered a chain that no longer exists.

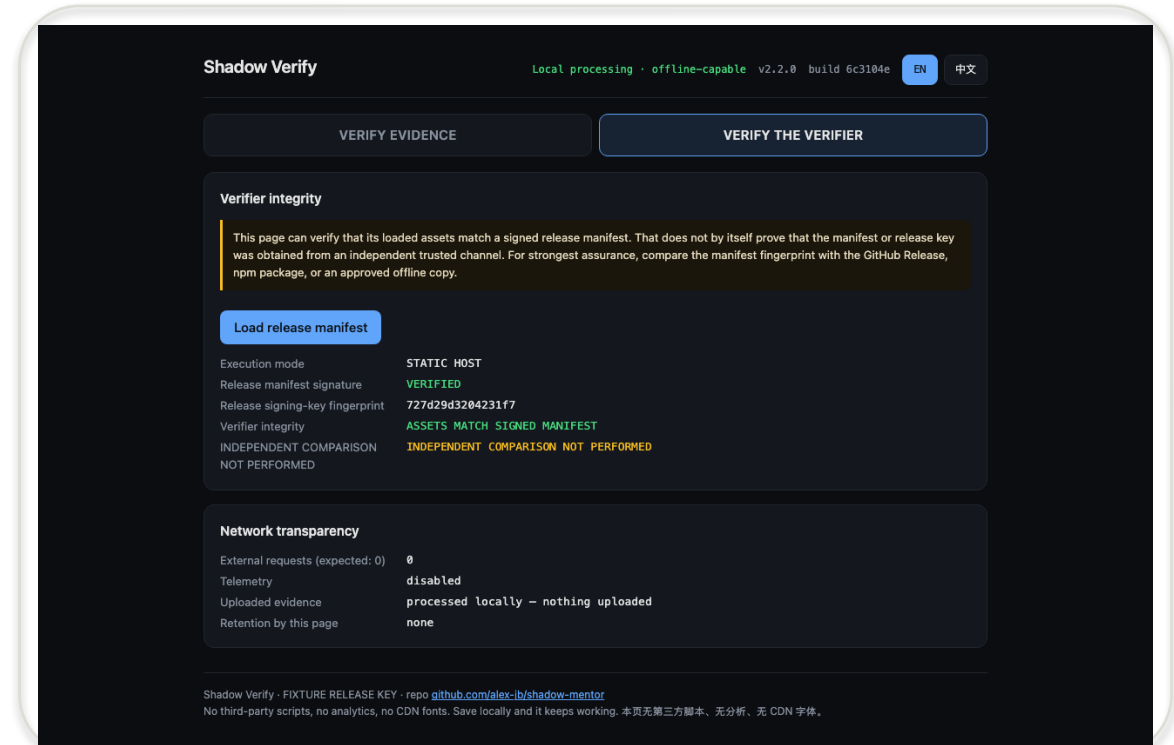
Even in failure, integrity and **correctness stay separate** — it never says the conclusion was right.

Can you trust the verifier page itself?

It carries a signed fixture manifest — **ASSETS MATCH SIGNED MANIFEST**.

Real trust needs an out-of-band check: asset hashes vs. a manifest from a separate channel, and the release-key fingerprint compared independently — **INDEPENDENT COMPARISON NOT PERFORMED**.

A page hashing itself is **not trust**.



Real render — Verify-the-Verifier (fixture manifest)

Current signing is **fixture, not production**.

3D for sequence, provenance, and tamper replay

Unity Shadow Lens + Three.js replay

- Three workspaces; provenance audit arc; Verify / Tamper / Reset
- Head-directed focus (gaze hover)
- A precise 2D fallback always remains
- Android mock APK built — 24.4 MB, hashed

Stated honestly

- ✗ NOT eye tracking — hover only
- ✗ No RGB capture
- ✗ No 6DoF on the mock build
 - Android APK is built, **not device-validated**
 - Beam Pro / XREAL validation pending
 - No user study yet — 3D benefit unproven

What is real today, measured honestly

1,824 / 1,827

host tests pass
(3 skipped · 0 failed)

Chromium 149

browser verifier
EN + 简体中文 · 0 external

24.4 MB APK

Android mock built
(not device-validated)

No user study

3D comprehension
not yet proven (RQ4)

Browser acceptance run in real Chromium (Playwright): 8 flows, English + Simplified Chinese,
CSP with 0 external requests / 0 violations, offline verification — real screenshots, not mock-ups.

Implemented now vs. Capstone II

IMPLEMENTED NOW

- ✓ Deterministic evidence + hash chain
- ✓ Independent + offline verification
- ✓ Exact tamper localization
- ✓ Bilingual verifier (EN / zh-CN)
- ✓ Android mock build

PENDING — CAPSTONE II

- Beam Pro / XREAL device validation
- Shared Unity / Three.js scene contract
- User study — does 3D actually help?
- Semantic audit of ingested output
- Production signing · KMS / HSM

Fixture-signed · device-pending · no correctness claim — stated plainly, not oversold.

CONTRIBUTION

Shadow does not ask the auditor to trust the AI's answer.

It gives them evidence they can independently verify.

The sequence, the sources, the signature — and, if it was tampered with, the exact break.

Integrity, not correctness.

Thank you — questions welcome.